

AMR SERIES

MODEL

AMR-300

The AMR-300 moves material between stations, stores and production lines without operators, tracks, or magnetic tape. LiDAR-based natural navigation maps your plant as-is, the fleet manager dispatches jobs from your ERP or MES, and roller-top or lift-top modules match your load units. Dual safety LiDAR plus bumpers keep it safe around people and forklifts.

Autonomous	LiDAR SLAM	Fleet Manager	Safety Rated
Payload 300 kg	Natural navigation, no tape / reflectors	ERP / MES integration, opportunity charging	Dual safety LiDAR + bumpers

01 SPECIFICATIONS

MECHANICAL

TYPE	AGV / Autonomous Mobile Robot
PAYLOAD	300 kg
NAVIGATION	Natural navigation (LiDAR SLAM)
DOCKING ACCURACY	± 10 mm

ENVIRONMENT / CONTROL

TOP MODULES	Roller-top / Lift-top / Custom
FLEET MANAGEMENT	Texsonics fleet manager, ERP/MES integration
CHARGING	Opportunity charging, auto-dock
SAFETY	Dual safety LiDAR + bumpers

02 PRODUCT



AMR-300 · AMR SERIES

03 WORKING RANGE

MOTION — GROUP A

TRAVEL SPEED	≤ 1.5 m/s
RUNTIME	~8 h per charge
MIN AISLE WIDTH	900 mm

MOTION — GROUP B

GRADEABILITY	≤ 5% incline
TURNING FOOTPRINT	Turns in place (differential drive)

04 OVERVIEW

INTRALOGISTICS
THAT DRIVES ITSELF
— NO TAPE, NO
TRACKS, NO
OPERATOR.

The AMR-300 moves material across your plant autonomously using LiDAR-based natural navigation. Jobs are dispatched from the Texsonics fleet manager and integrated with your ERP or MES.

01 NATURAL NAVIGATION

LiDAR SLAM maps your facility as-is. No floor markers, no reflectors, no infrastructure to install.

02 FLEET DISPATCH

Central fleet manager routes jobs to the nearest available unit and hands off work between shifts, cells and lines.

03 SAFETY BY DESIGN

Dual safety-rated LiDAR plus bumpers keep the AMR safe around operators, forklifts and mixed traffic.

05 CONTROL & INTEGRATION

CONTROLLER

Texsonics RC series
in-house motion controller,
dual-kernel real-time OS

TEACH PENDANT

11.5" touch tablet
drag-teaching, Wi-Fi,
on-device offline
programming

FIELDBUS

**EtherCAT / Modbus
TCP**
industrial fieldbus,
dual-protocol support

SOFTWARE

TEXCAM offline CAM
toolpath generation, digital
twin simulation, vision

06 APPLICATIONS

Autonomous material movement across production, warehousing and finished-goods flows — dispatched from the Texsonics fleet manager.

Line-side Delivery	Warehouse Transfer	WIP Movement	Finished Goods Transport		
Machine Tending	CNC Loading	Palletizing	Assembly	Packaging	Vision Inspection

07

BUILT FOR THE OPERATOR

- Live 3D position display
- Online command execution
- Position editor
- Variable monitor
- I/O monitor
- Error diagnostics
- Dynamic teach-in
- Digital twin simulation

08

SERVICE & SUPPORT

Made in Coimbatore. Supported from Coimbatore.

Every AMR-300 is engineered, built and calibrated in our Coimbatore facility. Field service, spares and applications engineering ship from the same address — no import lead times, no time-zone escalations.

STANDARD WARRANTY

12 months on the mechanical assembly, 12 months on the controller.

APPLICATIONS ENGINEERING

Cell design, cycle-time studies and end-of-arm tooling by our in-house team, included on every project.

SPARES & FIELD SERVICE

Direct-to-plant spares from Coimbatore, with pan-India field service engineers on 24-hour dispatch.

09

READY TO INTEGRATE

Talk to an engineer about integrating the AMR-300 into your line.

Applications engineering, cycle-time sizing, and turnkey cell design are included on every project.

CONTACT

Dharmar R

Managing Director

+91 94426 24304

dharmar@texsonics.net

www.texsonics.net

SPECIFICATIONS AND VISUALS MAY CHANGE WITHOUT NOTICE AS PART OF ONGOING PRODUCT IMPROVEMENTS. IMAGES ARE FOR REFERENCE ONLY.
UNAUTHORIZED REPRODUCTION OF THIS CATALOGUE IS PROHIBITED WITHOUT WRITTEN PERMISSION FROM TEXSONICS SYSTEMS INDIA PRIVATE LIMITED.

AMR-300

Texsonics Systems India Private Limited

1/6-1, Keerakaran Thottam, Keeranatham, Coimbatore 641035 • Ph: +91 94426 24304

AMR SERIES

AUTONOMOUS • PAYLOAD 300 kg • REACH Fleet-wide

dharmar@texsonics.net • www.texsonics.net